

REAL TIME MONITORING AND EVALUATION SOFTWARE FOR APPLICATION RECEIVED IN FIRE DEPARTMENT RELATING TO INSPECTIONS, FOLLOW-UPS, ISSUE OF NOCS.

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ABSTRACT

The Real-Time Monitoring and Evaluation Software for Fire Department Applications is a digital platform designed to streamline and automate the handling of applications submitted for fire safety inspections, follow-ups, and issuance of No Objection Certificates (NOCs). The system ensures transparency, efficiency, and timely processing of requests received by the fire department. This software enables users to submit applications online, attach required documents, and track the status of their request in real time. The system notifies officers about new applications, schedules inspections, records inspection outcomes, and generates automated progress updates for applicants. Follow-up reminders and pending case alerts help the department reduce delays and improve workflow efficiency. All application data is securely stored and categorized, allowing authorized personnel to review, approve, reject, or request additional information quickly. The dashboard provides visual insights through charts, reports, and timelines, offering management an overview of department performance, workload distribution, and NOC approval statistics. By integrating inspection modules, document management, and automated communication, the software ensures faster NOC clearance, enhances coordination among field officers and administrators, and promotes a transparent, accountable, and citizen-friendly service process.

I. INTRODUCTION

The Fire Department plays a vital role in ensuring the safety of citizens by inspecting buildings, verifying fire precautions, and issuing No Objection Certificates (NOCs). Traditionally, applications for inspection and NOC approval are processed manually, which often leads to delays, miscommunication, and difficulty in tracking application status.

To overcome these challenges, a Real-Time Monitoring and Evaluation software system is proposed. This system digitizes the entire workflow—application submission, document verification, inspection scheduling, status tracking, and approval process.

Applicants can submit requests online and receive live updates without visiting the department physically. Officers can review applications instantly, assign inspection duties, upload reports, and approve or reject requests based on compliance.

The system improves transparency, reduces paperwork, and enhances service delivery speed. Real-time dashboards allow administrators to monitor pending tasks and department performance. Overall, the proposed solution supports a systematic and efficient approach to public safety management through technology-driven automation.

II. LITERATURE SURVEY

1) 2018 / SMITH ET AL,

SMART WEARABLES FOR PERSONAL SAFETY.

A smart wearable system for personal safety using Bluetooth-based proximity sensing. It alerts users when they enter predefined danger zones. The system works effectively but is limited by short Bluetooth range and occasional false alerts.

2) IOT BASED SMOKE DETECTION AND TRANSISTOR.

An IoT-based smoke detection system uses gas or smoke sensors along with a transistor circuit to sense hazardous smoke levels. When smoke is detected, the signal is processed and sent to a microcontroller, which triggers alerts or notifications. The system provides automated monitoring and enhances safety through real-time detection and reporting.

3) OUTPUT FROM VARIOUS SENSORS IN ADC.

Sensors generate analog signals based on environmental changes such as smoke, flame, or temperature. These analog signals are fed into an ADC (Analog to Digital Converter) to convert them into digital values. The digital output is then processed by a microcontroller for real-time monitoring and decision-making.

4) FIRE SMOKE DETECTION SYSTEM USING SMS AND GSM SYSTEM

A fire smoke detection system using SMS and GSM technology identifies smoke levels through sensors and sends alerts instantly. Once smoke is detected, the GSM module transmits warning messages to the user via SMS. This enables real-time notification and quick emergency response even from remote locations.

III. ADVANTAGES

- **Real-time Tracking:** The system continuously monitors the status of fire safety applications. Users can check progress anytime without visiting the office. This improves transparency and reduces information delays.
- **Faster Application Processing:** Digital workflow reduces paperwork and manual verification. Officials can review, approve, or reject applications instantly. This speeds up inspection and NOC issuance.
- **Automated Notifications & Alerts:** Applicants receive SMS/Email updates at every stage. Pending tasks are highlighted for officers to avoid delay. This ensures timely communication and follow-up.
- **Efficient Inspection Scheduling:** Inspection dates can be assigned and updated digitally. Field officers get real-time alerts and location details. This improves coordination and timely completion of inspections.
- **Centralized Data Storage:** All documents, reports, and applications are stored securely. Data can be accessed anytime for review or analysis. Reduces risk of lost files and duplication.
- **Reduced Human Error:** Automation minimizes manual entry mistakes. Digital validation ensures accurate document submission. Reduces confusion, misplacement, and processing errors.
- **Performance Analytics & Dashboard:** Admin dashboard generates statistics, trends, and graphs. Authorities can evaluate workload, pending cases, and efficiency. Helps in better decision-making and resource management.
- **Improved Public Experience:** Citizens can apply from home and get approvals faster. No long queues, offline visits, or repeated follow-ups required. Creates a more user-friendly and trust-based system.

IV. DISADVANTAGES

- **Dependence on Internet Connectivity:**

The system requires a stable and continuous internet connection for accessing and updating applications. In areas with weak networks, users may face delays while submitting forms or checking status. This reduces the efficiency of real-time monitoring.

- **Technical Issues and System Downtime**

Software bugs, server failures, or maintenance periods can interrupt the entire workflow. During downtime, inspections and approvals may be stalled. This affects the reliability of the system.

- **High Initial Implementation Cost**

Installing servers, developing software, and training employees demand high upfront investment. Small fire departments may find this cost challenging. This can slow down system adoption.

- Requirement of Digital Literacy

Some staff and citizens may not be familiar with online platforms. Lack of digital skills can lead to errors or delays in application handling. Training becomes necessary for smooth functioning.

- Data Privacy and Security Risks

Storing sensitive documents online increases vulnerability to cyber-attacks. Without strong encryption and security protocols, personal and inspection data may be exposed. This can affect user trust.

Problem Statement:

The fire department currently relies on manual processes for receiving applications, scheduling inspections, tracking follow-ups, and issuing NOCs, leading to delays and inefficiencies. Applicants have no clear way to monitor the status of their requests, resulting in repeated visits and communication gaps. Paper-based records are prone to misplacement, errors, and slow retrieval during verification. Officials face difficulty managing workload, prioritizing pending cases, and maintaining timely inspections. The absence of automated alerts and real-time updates further adds to processing delays. Therefore, a digital system is needed to streamline operations, improve transparency, and ensure faster and more accurate handling of fire safety applications.

Existing System:

The current fire department process for application handling is mostly manual and paper-based. Applicants must visit the office to submit forms, provide documents, and inquire about the status of their requests. Inspection scheduling is done through physical registers or phone calls, which often leads to delays and miscommunication. Officers manually store reports and follow-up records, making retrieval time-consuming. There is no real-time tracking system, so applicants remain unaware of progress. Lack of automated reminders and monitoring also causes pending cases to accumulate. Overall, the existing system is slow, inefficient, and lacks transparency.

Proposed Solution:

- Digital Application Processing

The system provides an online platform for submitting applications and uploading documents. This eliminates manual paperwork and speeds up verification.

- Real-Time Monitoring & Tracking

Applicants and officials can track application status instantly through dashboards. This improves transparency and reduces unnecessary follow-ups.

- Automated Inspection & Follow-Up Management

The system assigns inspection dates, sends reminders, and stores inspection reports digitally. This ensures timely completion of tasks and reduces delays.

- Smart Alerts & NOC Issuance

SMS/Email alerts notify users about approvals, rejections, or required actions. The system generates digital NOCs, improving accuracy and service efficiency.

Benefits of Proposed System:

- Faster Processing of Applications

Digital submission and automated workflow reduce manual delays, making the entire inspection and NOC approval process quicker.

- Improved Transparency

Applicants can track the real-time status of their application, reducing confusion and eliminating the need for repeated follow-ups.

- Enhanced Accuracy & Reduced Errors

Automation minimizes human errors in data entry, document handling, and status updates, ensuring more reliable results.

- Timely Inspections & Follow-Ups

Automated reminders and scheduling ensure that inspections and follow-up actions are completed on time without backlog.

- Better Communication

SMS/Email notifications keep both officials and applicants updated at every stage, improving overall coordination.

- Centralized Data Storage

Documents, inspection reports, and application details are stored securely, making retrieval and auditing quick and easy.

- Increased Public Satisfaction

Citizens save time and effort with faster services, easy tracking, and reduced need to visit the fire department.

- Strong Accountability & Monitoring

Dashboards and logs help officials monitor pending tasks, workloads, and officer performance, ensuring better governance.

V. METHODOLOGIES

a) Front-end methodology :

HTML (HyperText Markup Language) is used to design and structure all the web pages of the Real-Time Monitoring and Evaluation System. It provides the base layout for displaying application forms, dashboards, inspection details, and NOC status updates. HTML ensures that the interface remains clear, well-structured, and easy for applicants and officers to navigate.

Using various HTML tags, the system creates organized sections such as input fields, buttons, tables, and status indicators that help users interact with the platform smoothly. HTML is lightweight, works on all browsers, and integrates seamlessly with CSS and JavaScript to create a responsive and interactive user experience.

In this project, HTML is mainly used to create:

Application submission forms

Applicant and officer login pages

Dashboard for tracking status

Screens for inspection reports & NOC display

- Cascading Style Sheets (CSS).

CSS (Cascading Style Sheets) is used to enhance the visual appearance, usability, and responsiveness of the system's user interface. It defines the colors, fonts, layouts, and overall style of the web pages, making the platform clean, professional, and easy to understand for both applicants and fire department staff.

CSS helps maintain a consistent design across all pages such as forms, tables, navigation panels, and response cards. It improves readability and keeps the interface organized. With responsive design techniques like media queries, CSS ensures that the system works smoothly across desktops, tablets, and mobile devices allowing users to track their applications anytime.

Animations, hover effects, and transition styles are used to make buttons, alerts, and status indicators more interactive and intuitive. Overall, CSS plays a key role in improving user experience and helping the system appear modern and user-friendly.

- JavaScript

JavaScript adds interactivity and dynamic functionality to the Fire Department Monitoring System. It enables real-time updates, form validation, data fetching, and dashboard interactions without requiring full page reloads. JavaScript ensures that users can view live status updates for inspections, follow-ups, and NOC approvals.

It communicates with backend APIs to retrieve application data, inspection schedules, and alert messages instantly. JavaScript is also used to validate user inputs, prevent errors during form submission, and enhance overall accuracy. Officers can view assigned inspections, upload reports, and update status efficiently through JavaScript-powered components.

Features like dynamic tables, interactive charts, dropdown filters, and notification pop-ups make the system simpler and faster to use. JavaScript ensures that the entire workflow from application submission to NOC issuance feels responsive, smooth, and real-time

b) Back-end:

- **Flask Backend Framework**

Flask is a lightweight and powerful Python-based backend web framework used to build scalable and secure web applications. It enables smooth communication between the frontend interface and the database while handling all server-side logic efficiently. Flask is well suited for government and departmental applications because of its simplicity, flexibility, and strong integration with databases.

In this project, Flask is used as the backend framework to manage fire department applications related to inspection, follow-ups, and issuance of No Objection Certificates (NOCs). It handles application submissions, inspection reports, officer updates, and status tracking in a structured and secure manner. Flask processes user requests, performs backend operations, and sends real-time responses to the frontend dashboard.

The backend uses SQLite database to store application data securely and ensures reliable data handling for applicants, officers, and administrators. Flask's REST API structure allows smooth data exchange between different system modules. Since the application runs on a server environment, it reduces manual paperwork and improves response time for fire department operations.

- Key Features of Flask Backend Used in This Project
- Database Management (SQLite)
 - Stores fire safety application forms, applicant details, inspection reports, and NOC status.
 - Maintains structured records for follow-ups and historical tracking.
 - Ensures data consistency and secure storage.
- RESTful API Services
 - Provides APIs for application submission, inspection updates, and NOC approval.
 - Enables smooth interaction between frontend dashboards and backend logic.
 - Ensures fast response and efficient data processing.
- Role-Based Operations
 - Supports different user roles such as applicant, inspection officer, and admin.
 - Controls access to system features based on user responsibility.
 - Prevents unauthorized data access.
- Real-Time Status Updates
 - Application status updates (Pending, Inspected, NOC Issued) are reflected instantly.
 - Officers and administrators can update records without delays.
 - Applicants can track their application progress efficiently.

- Security & Reliability
- Backend logic validates requests before database operations.
- Prevents data duplication and unauthorized changes.
- Ensures stable performance for departmental use.
- Scalability & Maintainability
- Flask architecture allows easy future enhancements.
- Can be extended with authentication, PDF NOC generation, and cloud deployment.
- Suitable for handling increasing numbers of applications.

VI. WORK FLOW

A) User Registration & Login

Applicants, fire officers, and administrators access the system through a secure login interface. Users enter their credentials to log in, and the system verifies their role (Applicant / Officer / Admin) before granting access to relevant features.

B) Application Submission

Applicants submit fire safety applications through the system by entering required details such as:

- Applicant name
- Building address
- Fire safety information

Once submitted, the application is stored securely in the backend database with a “Pending” status.

C) Application Review by Admin

Administrators log into the system and view all submitted applications through the dashboard. They verify application details and assign applications to inspection officers for further processing.

D) Inspection Scheduling & Execution

Fire officers receive assigned inspection requests.

Officers conduct on-site inspections and record findings, remarks, and inspection results using the system. Inspection details are submitted through the backend and linked to the respective application.

E) Inspection Data Processing

The backend processes inspection data to ensure completeness and accuracy.

Based on inspection remarks, the application status is updated to “Inspected” or marked for follow-up if corrections are required.

F) Application Status Update

Each application progresses through defined stages:

- Pending
- Inspected
- Approved / Rejected
- NOC Issued

Status changes are reflected immediately in the system, allowing applicants to track progress in real time.

G) NOC Generation & Issuance

After successful inspection and approval, administrators issue the No Objection Certificate (NOC). The application status is updated to "NOC Issued", and the final approval record is stored securely in the database.

H) Data Storage & Record Management

All application data, inspection reports, officer remarks, and NOC records are stored in the backend database. This ensures long-term record maintenance, easy retrieval, and transparency for departmental audits.

I) Reporting & Monitoring

Administrators can view reports related to:

- Number of applications received
- Inspection status
- Pending and approved NOCs

This helps in monitoring departmental performance and improving response time.

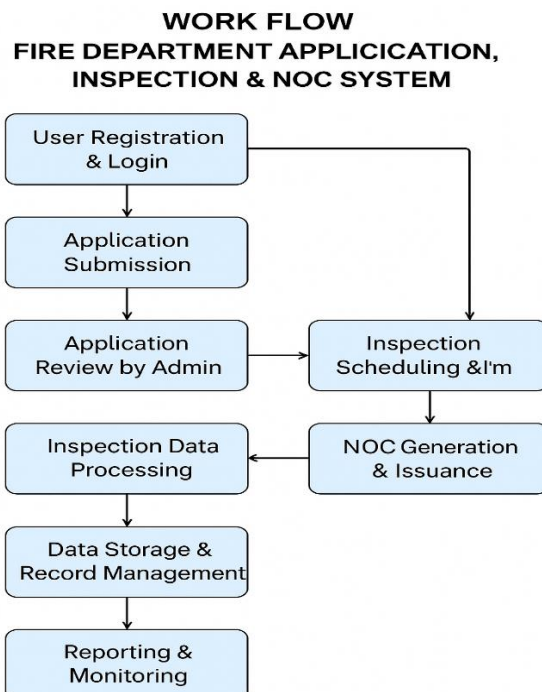
J) Data Security & Access Control

The system ensures data privacy by:

- Allowing access based on user roles
- Preventing unauthorized data modification
- Securing sensitive applicant and inspection information

Only authorized users can perform specific actions within the system.

Flow Diagram:



Modules:

[1] USER LOGIN METHODOLOGY

When a user accesses the website, the Login Page is displayed. After entering valid credentials, the system redirects the user to the User Dashboard.

The User Dashboard contains the following sections:

- Message / Notifications
- Fire incident application submission
- View applied fire inspection or NOC requests
- Incident history
- Status tracking of applications
- Gallery and report view

This dashboard allows users to interact with the fire department digitally and track the progress of their applications.

[2] AUTHENTICATION

- The system uses secure authentication and authorization mechanisms to ensure that only authorized users can access the platform.
- User identification is performed using login credentials stored in the database (noc_system.db).
- Access control is implemented based on user roles and permissions.

There are two types of user roles in the system:

1. User / Applicant – Can submit fire incident reports, apply for inspection or NOC, and view application status.
2. Administrator / Fire Officer – Can view all applications, analyze incidents, manage users, and monitor system activity.

[3] CREATE INCIDENT / APPLICATION

- Users can submit fire incident details or inspection/NOC applications using the application form available on the dashboard.
- The submitted data includes:
 - Applicant details
 - Location of incident or building
 - Incident description
 - Uploaded images or documents
- Once submitted, the application is stored in the database and forwarded to the Administrator Dashboard for review and action.

[4] VIEW APPLICATION / INCIDENT STATUS

- Users can view the status of their submitted applications by clicking on the View Application / View Fire Incident option.
- The system displays:
 - Application ID
 - Submission date
 - Location
 - Current status (Pending / Approved / Rejected)
 - Remarks from fire officers

This ensures transparency and keeps applicants informed throughout the process.

[5] ADMIN MODULE

- Administrators can log in through the Admin Login Page.
- The Admin Dashboard provides features such as:
 - Viewing all fire incident reports
 - Reviewing inspection and NOC applications
 - AI-based incident monitoring and analysis
 - Managing user records
 - Admins can update application status and provide remarks after inspection.

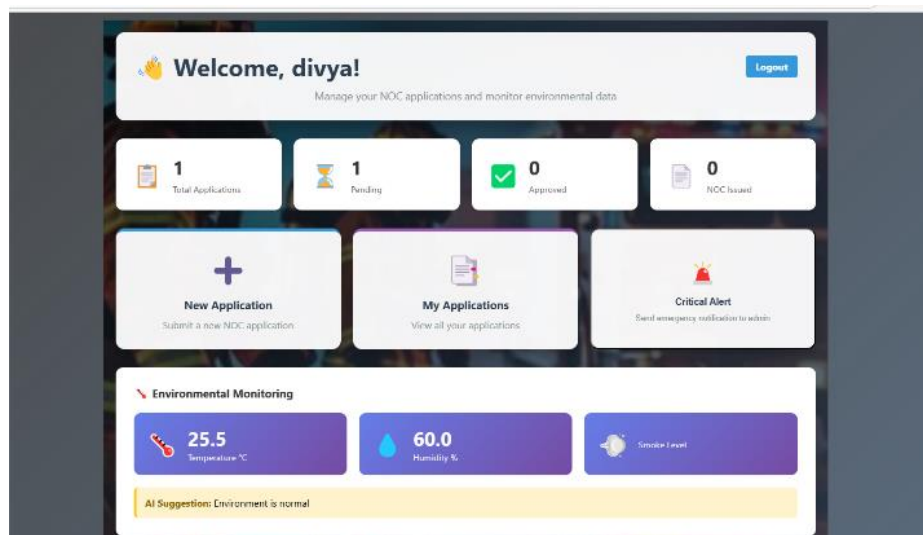
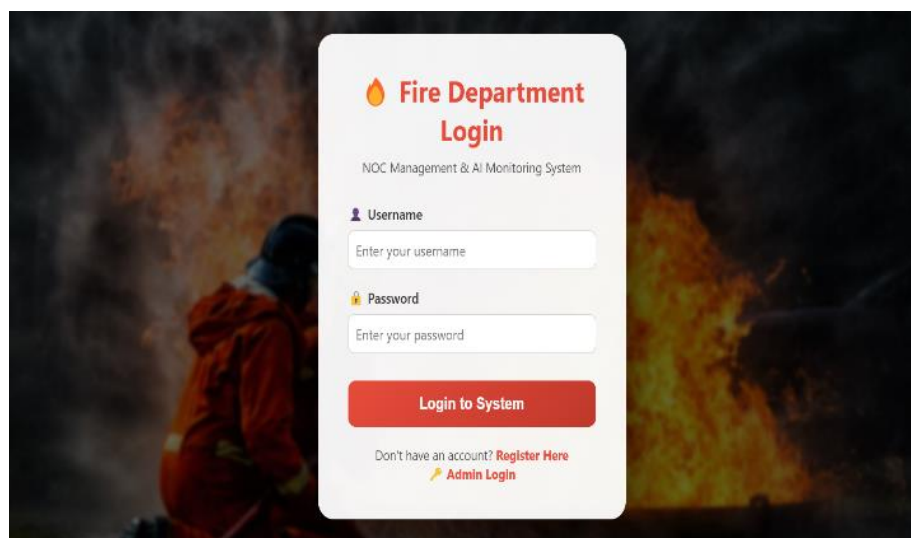
[6] AI & IoT MONITORING MODULE

The system includes AI-based monitoring pages to assist in analyzing fire-related data.

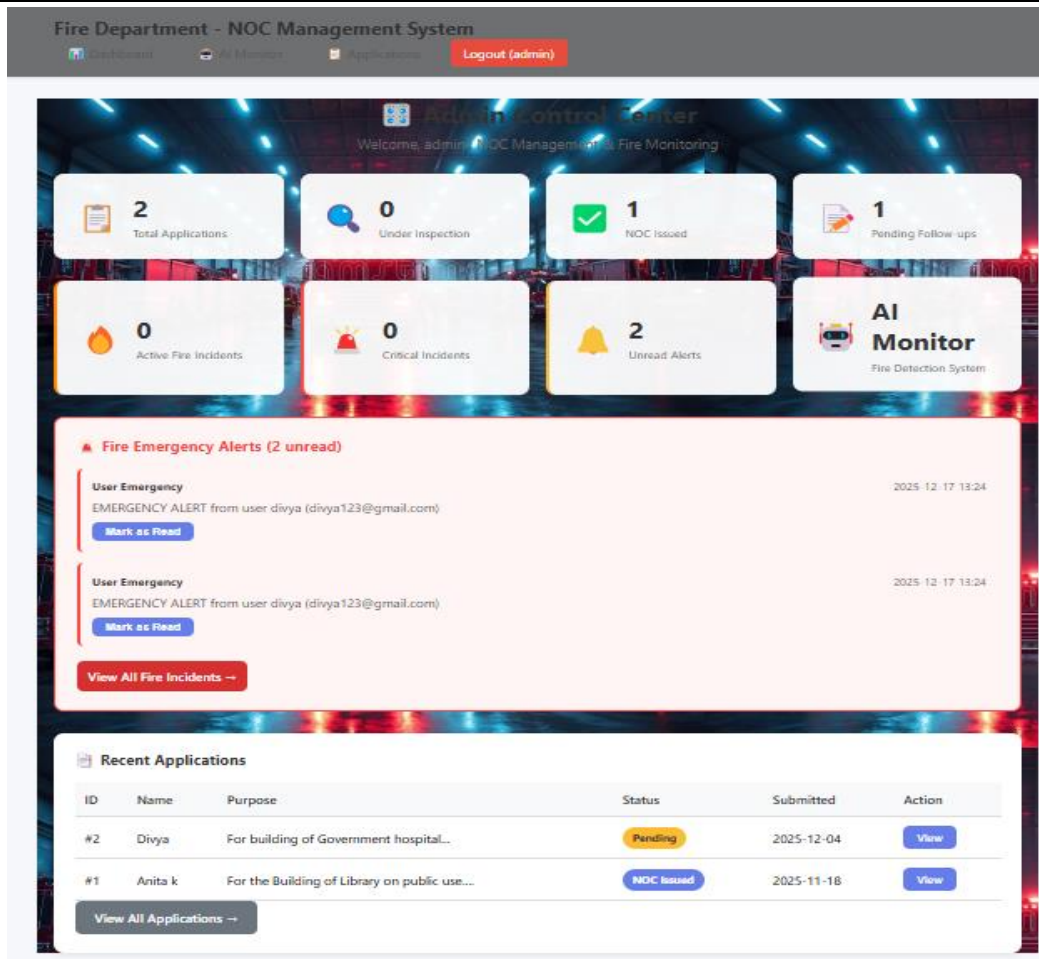
This module helps in identifying risk patterns, monitoring incidents, and improving decision-making efficiency in the fire department.

Login Page

The **Login Page** serves as the secure entry point for both users and administrators. Only authenticated users are allowed access to internal system modules, ensuring **data security and system integrity**.



User Dashboard.



Admin Dashboard.

VII. CONCLUSION

The Real-Time Monitoring and Evaluation System for Fire Department Applications provides an efficient digital platform for managing fire incidents, inspections, and NOC approvals. By replacing manual processes with an online system, the project improves accuracy, transparency, and response time.

The integration of AI-based monitoring and centralized dashboards enhances operational efficiency and supports better decision-making for fire department officials. Overall, this system acts as a reliable solution for modernizing fire department application management and incident monitoring.

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